



December 2025

(For BTY)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following cells is a primary effector of innate immunity?
A) Neutrophil
B) B lymphocyte
C) Plasma cell
D) T lymphocyte
- Which of the following cells is both a phagocyte and an antigen-presenting cell (APC)?
A) Eosinophil
B) Macrophage
C) NK cell
D) Basophil
- Dendritic cells are important because they:
A) Produce antibodies
B) Release histamine
C) Present antigens to T cells
D) Kill infected cells
- Which type of immunity is induced by vaccines?
A) Innate
B) Passive
C) Non-specific
D) Adaptive
- Recombinant vaccines are based on:
A) Whole killed organisms
B) Live attenuated organisms
C) Polysaccharides only
D) Specific genes encoding antigens
- Adjuvants in vaccines:
A) Enhance immune response
B) Decrease immune response
C) Destroy pathogens
D) Act as antibiotics
- The main antibody class responsible for allergic reactions is:
A) IgG
B) IgA
C) IgE
D) IgM
- Conventional whole organism vaccines include:
A) Live attenuated and killed vaccines
B) Recombinant protein vaccines
C) Subunit vaccines
D) DNA vaccines
- Type I hypersensitivity is primarily mediated by:
A) IgG
B) IgA
C) IgM
D) IgE
- Macrophage activation in tuberculosis leads to:
A) IgE production
B) Granuloma formation
C) NK cell activation
D) Immediate hypersensitivity
- AIDS is characterized by:
A) Overproduction of antibodies
B) Loss of CD4+ T cells
C) Granuloma formation
D) Increased neutrophils

12. Autografts are:
 A) Transplants within the same individual
 B) Transplants between individuals of the same species
 C) Transplants from animals to humans
 D) Transplants between identical twins
13. Immune tolerance refers to:
 A) Selective unresponsiveness to self-antigens
 B) Overactive immune response
 C) Hypersensitivity to allergens
 D) Production of autoantibodies
14. Stem cells in transplantation are used to:
 A) Replace damaged tissues and regenerate immune cell
 B) Produce antibodies
 C) Mediate hypersensitivity reactions
 D) Activate granulomas
15. Immunosuppressive drugs such as cyclosporine function by:
 A) Stimulating B cells
 B) Increasing antibody production
 C) Enhancing NK cell activity
 D) Inhibiting T cell activation
16. The interaction between soluble antigen and soluble antibody forming a visible lattice is called:
 A) Agglutination
 B) Precipitation
 C) ELISA
 D) Immunofluorescence
17. In double immunodiffusion (Ouchterlony method), both antigen and antibody:
 A) Diffuse towards each other
 B) Remain fixed
 C) Diffuse in the same direction
 D) Do not diffuse
18. Immunofluorescence is most useful in detecting:
 A) Soluble antigens only
 B) Cellular or tissue-bound antigens
 C) Radioactive antigen
 D) Serum antibody concentration
19. Chemiluminescent immunoassay is based on detection of:
 A) Light emitted during chemical reactions
 B) Fluorescence under UV light
 C) Radioactive decay
 D) Precipitation lines
20. A major limitation of RIA is:
 A) Low sensitivity
 B) Hazard of radioactive isotopes
 C) Lack of specificity
 D) High cost of antibodies

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the functions and interplay of major innate immune cells, including macrophages, neutrophils, and natural killer (NK) cells.	8	L*1	1	1
	b) Define antigens and provide a detailed discussion on the general characteristics that define an antigen.	8	1	1	1
2.	a) Discuss the roles and immunological significance of haptens and adjuvants in modulating immune responses.	8	1	1	1
	b) Write brief notes on: i) DNA vaccines, and ii) the thymus.	8	1	2	1
3.	a) With a clear, labeled diagram, describe the detailed structure of an antibody molecule.	8	1	2	1
	b) Provide a detailed explanation of the endocytic pathway involved in antigen processing and presentation.	8	1	2	1
Unit – II					
4.	a) Explain Type I hypersensitivity in humans and analyze its impact on health.	8	1	3	1,4
	b) Describe the process of macrophage activation and granuloma formation, highlighting their immunological significance.	8	1	3	1,4

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|-------|--|---|---|---|-----|
| 5. a) | Discuss the immunological principles underlying allergic diseases. | 8 | 1 | 3 | 1,4 |
| b) | Explain the immunological mechanisms responsible for graft rejection. | 8 | 1 | 4 | 1,2 |
| 6. a) | With reference to Type II hypersensitivity, explain its mechanism and provide relevant examples. | 8 | 1 | 4 | 1,2 |
| b) | Evaluate the use of stem cells in transplantation, including clinical applications and ethical considerations. | 8 | 1 | 4 | 1,2 |

Unit – III

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|-------|--|---|---|---|-------|
| 7. a) | Discuss the principle, procedure, and applications of ELISA in immunological studies. | 8 | 1 | 5 | 1,3,4 |
| b) | Provide a detailed account of antigen–antibody interactions, highlighting factors that influence binding. | 8 | 1 | 5 | 1,3,4 |
| 8. a) | Using the precipitin reaction as an example, explain the method of immunoelectrophoresis and its significance. | 8 | 1 | 5 | 1,3,4 |
| b) | Give an overview of non-radioactive immunological detection methods, discussing their principles and advantages. | 8 | 1 | 5 | 1,3,4 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
